

2019 Ph H2 Q6

Section: Our Dynamic Universe

Topic: Blackbody Radiation — Wien's Law

- (a) The graph shows the blackbody radiation curve for a star at 5000 K. Sketch the curve for a star at 6000 K.
(b) Data for several stars is provided. Use the data to show that $\lambda_{\text{peak}} T = 2.9 \times 10^{-3} \text{ m K}$.

Worked solution

(a) At higher temperature (6000 K):

- Peak wavelength shifts to shorter λ (curve moves left).
- Peak is higher (more energy emitted overall).

Sketch: Draw a curve similar in shape, but with peak further left and higher than the 5000 K curve.

(b) Wien's Law states $\lambda_{\text{peak}} T = \text{constant}$.
Using data from the table:

- For $T = 7700 \text{ K}$, $\lambda = 3.76 \times 10^{-7} \text{ m}$, product = $2.90 \times 10^{-3} \text{ mK}$.
- For $T = 8500 \text{ K}$, $\lambda = 3.42 \times 10^{-7} \text{ m}$, product = $2.91 \times 10^{-3} \text{ mK}$.
- For $T = 9600 \text{ K}$, $\lambda = 3.01 \times 10^{-7} \text{ m}$, product = $2.89 \times 10^{-3} \text{ mK}$.
- For $T = 12\,000 \text{ K}$, $\lambda = 2.42 \times 10^{-7} \text{ m}$, product = $2.90 \times 10^{-3} \text{ mK}$.

All are approximately equal to $2.9 \times 10^{-3} \text{ mK}$.

Therefore, $\lambda_{\text{peak}} T \approx 2.9 \times 10^{-3} \text{ mK}$, as required.

Final answers

(a) Curve shifts left and higher for 6000 K.

(b) $\lambda_{\text{peak}} T \approx 2.9 \times 10^{-3} \text{ mK}$ (Wien's displacement law).

Revision tips

- Blackbody curve: higher T shifts peak to shorter wavelength.
- Peak also becomes higher (greater intensity).
- Wien's Law: $\lambda_{\text{peak}} T = \text{constant}$.
- Check by multiplying λ_{peak} and T for each star — result should be consistent.
- λ in metres, T in kelvin.